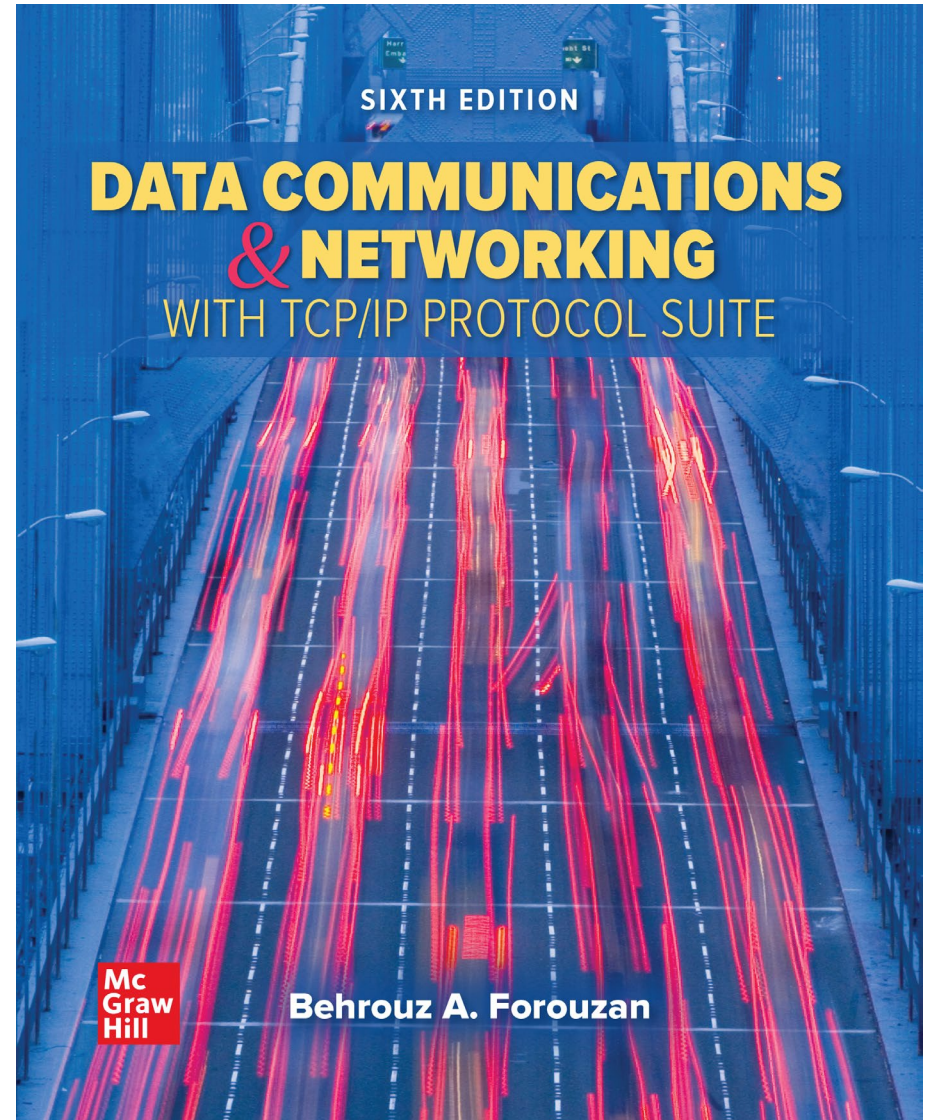


Chapter 07

Network Layer: Data Transfer

Data Communications and
Networking, With TCP/IP
protocol suite
Sixth Edition
Behrouz A. Forouzan



Chapter 7: Outline

7.1 Services

7.2 Packet Switching

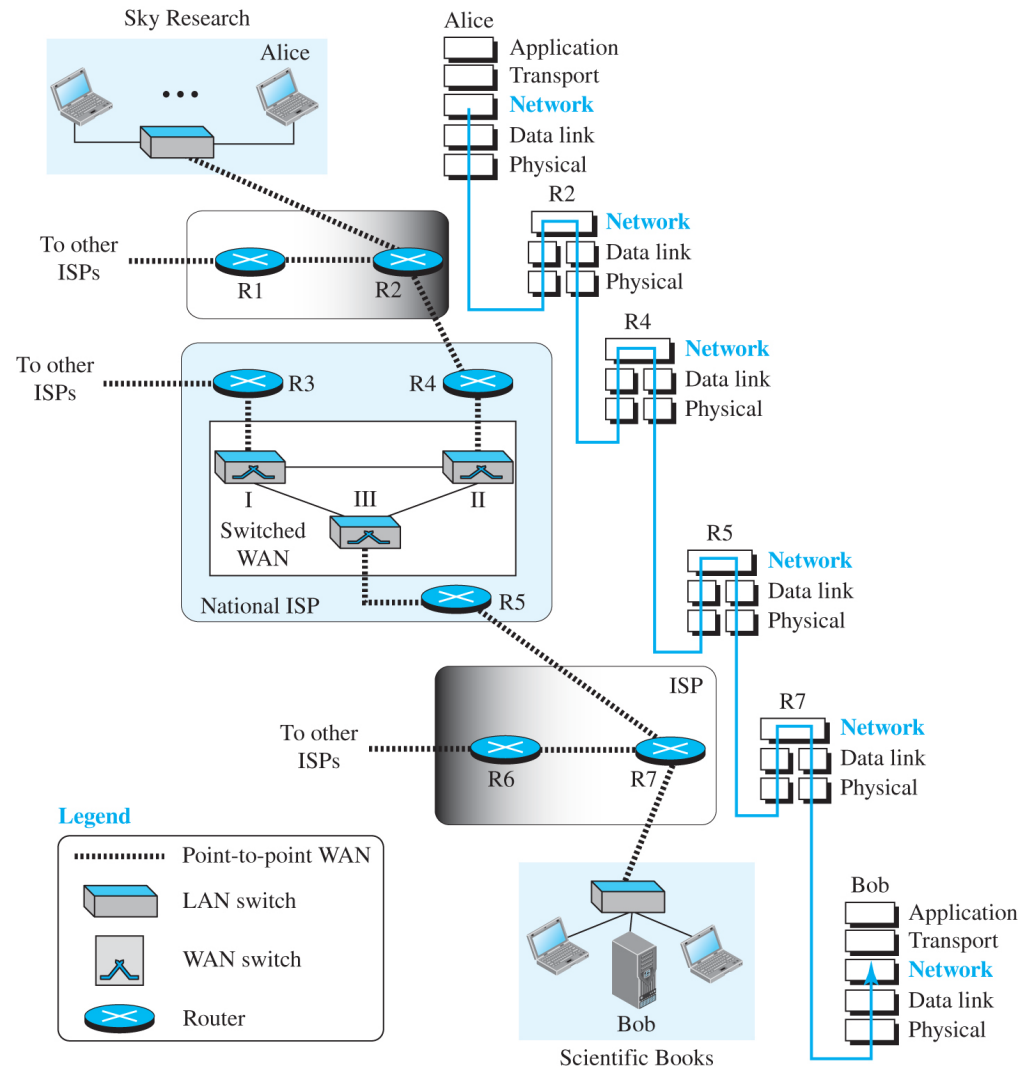
7.3 Performance

7.4 Internet Protocol V4

7.5 Internet Protocol V6

7.6 Transition from V4 To V6

Figure 7.1 Communication at the network layer



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7-1 SERVICES

We briefly discuss the services provided at the network layer.

7.1.1 Packetizing

The first duty of the network layer is definitely packetizing:

- *encapsulating the payload in a network-layer packet at the source and decapsulating the payload from the network-layer packet at the destination.*
- *In other words, one duty of the network layer is to carry a payload from the source to the destination without changing it or using it.*
- *The network layer is doing the service of a carrier such as the postal office, which is responsible for delivery of packages from a sender to a receiver without changing or using the contents.*

7.1.2 Routing

Other duties of the network layer, which are as important as the first, are routing and forwarding, which are directly related to each other.

7.1.3 Error Control

- *Although error control can be implemented in the network layer, the designers of the network layer in the Internet ignored this issue for the data being carried by the network layer.*
- *One reason for this decision is the fact that the packet in the network layer may be fragmented at each router, which makes error checking at this layer inefficient.*

7.1.4 Flow Control

- *Flow control regulates the amount of data a source can send without overwhelming the receiver.*
- *If the upper layer at the source computer produces data faster than the upper layer at the destination computer can consume it, the receiver will be overwhelmed with data.*
- *To control the flow of data, the receiver needs to send some feedback to the sender to inform the latter that it is overwhelmed with data.*

7.1.5 Congestion Control

- *Another issue in a network-layer protocol is congestion control.*
- *Congestion in the network layer is a situation in which too many datagrams are present in an area of the Internet.*
- *Congestion may occur if the number of datagrams sent by source computers is beyond the capacity of the network or routers.*

7.1.6 Quality of Service

- *As the Internet has allowed new applications such as multimedia communication (in particular real-time communication of audio and video), the quality of service (QoS) of the communication has become more and more important.*
- *The Internet has thrived by providing better quality of service to support these applications.*
- *However, to keep the network layer untouched, these provisions are mostly implemented in the upper layer.*

7.1.7 Security

- *Another issue related to communication at the network layer is security.*
- *Security was not a concern when the Internet was originally designed because it was used by a small number of users at universities for research activities; other people had no access to the Internet.*
- *The network layer was designed with no security provision. Today, however, security is a big concern.*
- *To provide security for a connectionless network layer, we need to have another virtual level that changes the connectionless service to a connection-oriented service.*

7-2 PACKET SWITCHING

- *From the discussion of routing and forwarding in the previous section, we infer that a kind of switching occurs at the network layer.*
- *A router, in fact, is a switch that creates a connection between an input port and an output port (or a set of output ports), just as an electrical switch connects the input to the output to let electricity flow.*

7.2.1 Datagram Approach

- *When the Internet started, to make it simple, the network layer was designed to provide a connectionless service in which the network-layer protocol treats each packet independently, with each packet having no relationship to any other packet.*
- *The idea was that the network layer is only responsible for delivery of packets from the source to the destination.*
- *In this approach, the packets in a message may or may not travel the same path to their destination.*

7.2.2 *Virtual-Circuit Approach*

- *In a connection-oriented service (also called virtual-circuit approach), there is a relationship between all packets belonging to a message.*
- *Before all datagrams in a message can be sent, a virtual connection should be set up to define the path for the datagrams.*
- *After connection setup, the datagrams can all follow the same path.*
- *In this type of service, not only must the packet contain the source and destination addresses, it must also contain a flow label, a virtual circuit identifier that defines the virtual path the packet should follow.*

7-3 PERFORMANCE

- *The upper-layer protocols that use the service of the network layer expect to receive an ideal service, but the network layer is not perfect.*
- *The performance of a network can be measured in terms of delay, throughput, and packet loss.*
- *Congestion control is an issue that can improve the performance.*

7.3.1 Delay

- *All of us expect instantaneous response from a network, but a packet, from its source to its destination, encounters delays.*
- *The delays in a network can be divided into four types: transmission delay, propagation delay, processing delay, and queuing delay.*
- *Let us first discuss each of these delay types and then show how to calculate a packet delay from the source to the destination.*

Transmission Delay₁

- *A source host or a router cannot send a packet instantaneously.*
- *A sender needs to put the bits in a packet on the line one by one.*
- *If the first bit of the packet is put on the line at time t_1 and the last bit is put on the line at time t_2 , transmission delay of the packet is $(t_2 - t_1)$.*
- *Definitely, the transmission delay is longer for a longer packet and shorter if the sender can transmit faster.*
- *In other words, the transmission delay is .*

$$\text{Delay}_{\text{tr}} = (\text{Packet length}) / (\text{Transmission rate})$$

Propagation Delay

- Propagation delay is the time it takes for a bit to travel from point A to point B in the transmission media.*
- The propagation delay for a packet-switched network depends on the propagation delay of each network (LAN or WAN).*
- The propagation delay depends on the propagation speed of the media, which is 3×10^8 meters/second in a vacuum and normally much less in a wired medium; it also depends on the distance of the link.*
- In other words, propagation delay is .*

$$\text{Delay}_{\text{pg}} = (\text{Distance}) / (\text{Propagation speed})$$

Processing Delay

- *The processing delay is the time required for a router or a destination host to*
 - *receive a packet from its input port,*
 - *remove the header,*
 - *perform an error detection procedure,*
 - *and deliver the packet to the output port (in the case of a router) or deliver the packet to the upper-layer protocol (in the case of the destination host).*
- *The processing delay may be different for each packet, but normally is calculated as an average.*

Delay_{pr} = Time required to process a packet

Total Delay

- *Assuming equal delays for the sender, routers, and receiver, the total delay (source-to-destination delay) a packet encounters can be calculated if we know the number of routers, n , in the whole path.*

Delay_{qu} = The time a packet waits in queues

Transmission Delay₂

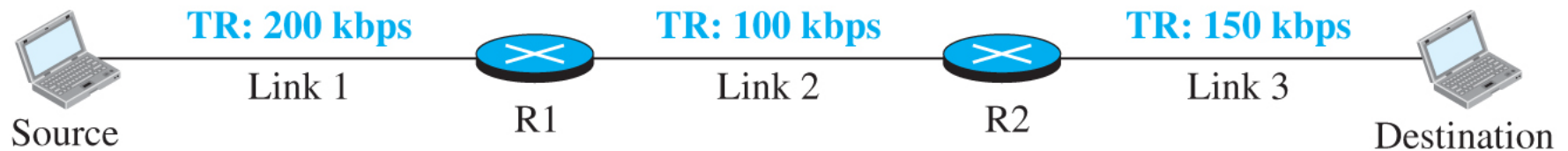
- *A source host or a router cannot send a packet instantaneously.*
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- *Definitely, the transmission delay is longer for a longer packet and shorter if the sender can transmit faster.*
- *In other words, the transmission delay is:*

$$\text{Delay}_{\text{tr}} = (\text{Packet length}) / (\text{Transmission rate})$$

7.3.2 Throughput

- *Throughput at any point in a network is defined as the number of bits passing through the point in a second, which is actually the transmission rate of data at that point.*
- *In a path from source to destination, a packet may pass through several links (networks), each with a different transmission rate.*
- *How, then, can we determine the throughput of the whole path?*
- *To see the situation, assume that we have three links, each with a different transmission rate, as shown in Figure 7.2.*

Figure 7.2 Throughput in a path with three links in a series



a. A path through three links

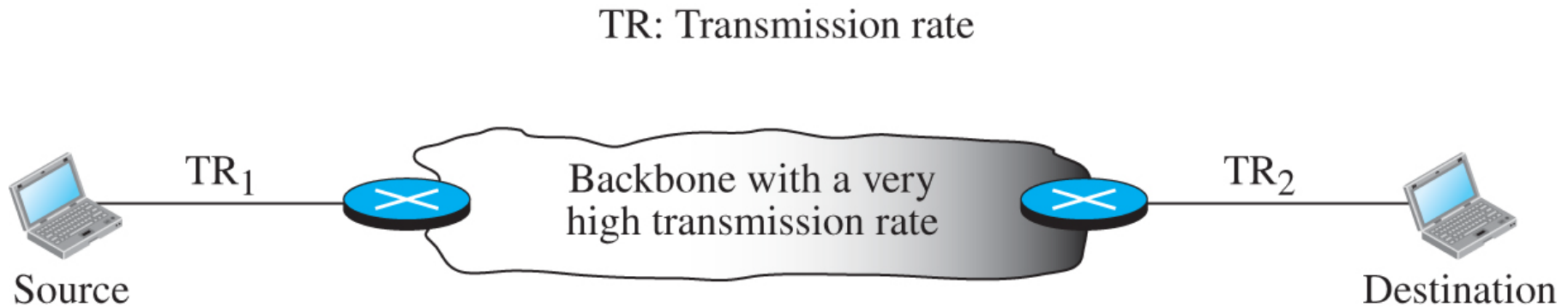
TR: Transmission rate



b. Simulation using pipes

[Access the text alternative for slide images.](#)

Figure 7.3 A path through the Internet backbone



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7.3.3 Packet Loss

- *Another issue that severely affects the performance of communication is the number of packets lost during transmission.*
- *When a router receives a packet while processing another packet, the received packet needs to be stored in the input buffer waiting for its turn.*
- *A router, however, has an input buffer with a limited size.*
- *A time may come when the buffer is full and the next packet needs to be dropped.*
- *The effect of packet loss on the Internet network layer is that the packet needs to be resent, which in turn may create overflow and cause more packet loss.*

7.3.4 Congestion Control

- *Congestion control is a mechanism for improving performance.*
- *Although congestion at the network layer is not explicitly addressed in the Internet model, the study of congestion at this layer may help us to better understand the cause of congestion at the transport layer and find possible remedies to be used at the network layer.*
- *Congestion at the network layer is related to two issues, throughput and delay, which we discussed in the previous section.*

7-4 INTERNET PROTOCOL VERSION 4

- *The network layer in the Internet has gone through several versions, but only two versions have survived: IP Version 4 (IPv4) and IP Version 6 (IPv6).*
- *Although IPv4 is almost depleted, we discuss it because there are still some areas that use this version and also because it is the foundation for IPv6.*

7.4.1 IPv4 Addressing

- *The identifier used in the IP layer of the TCP/IP protocol suite to identify the connection of each device to the Internet is called the Internet address or IP address.*
- *An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a host or a router to the Internet.*
- *The IP address is the address of the connection, not the host or the router, because if the device is moved to another network, the IP address may be changed.*

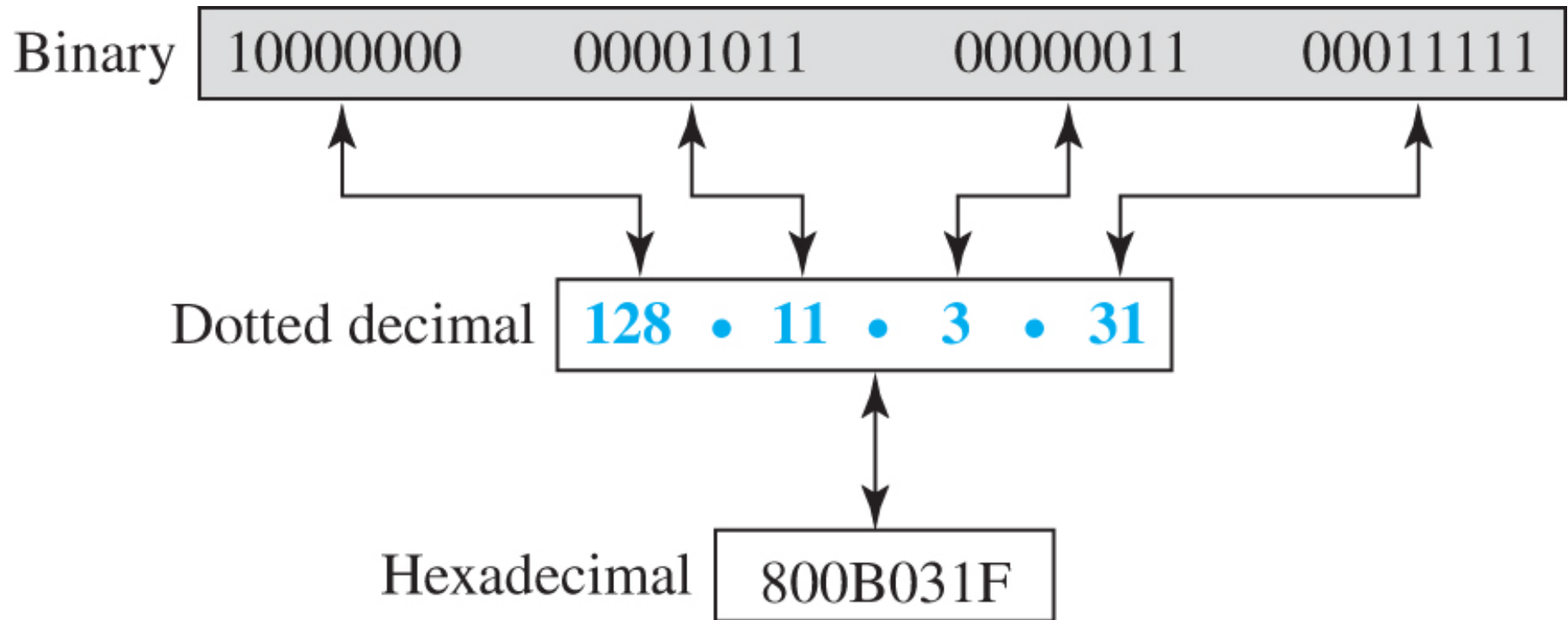
Address Space₁

- *A protocol like IPv4 that defines addresses has an address space.*
- *An address space is the total number of addresses used by the protocol.*
- *If a protocol uses b bits to define an address, the address space is 2^b because each bit can have two different values (0 or 1).*
- *IPv4 uses 32-bit addresses, which means that the address space is 2^{32} or 4,294,967,296 (more than four billion).*
- *If there were no restrictions, more than 4 billion devices could be connected to the Internet.*

Notation

There are three common notations to show an IPv4 address: binary notation (base 2), dotted-decimal notation (base 256), and hexadecimal notation (base 16).

Figure 7.4 Three different notations in IPv4 addressing

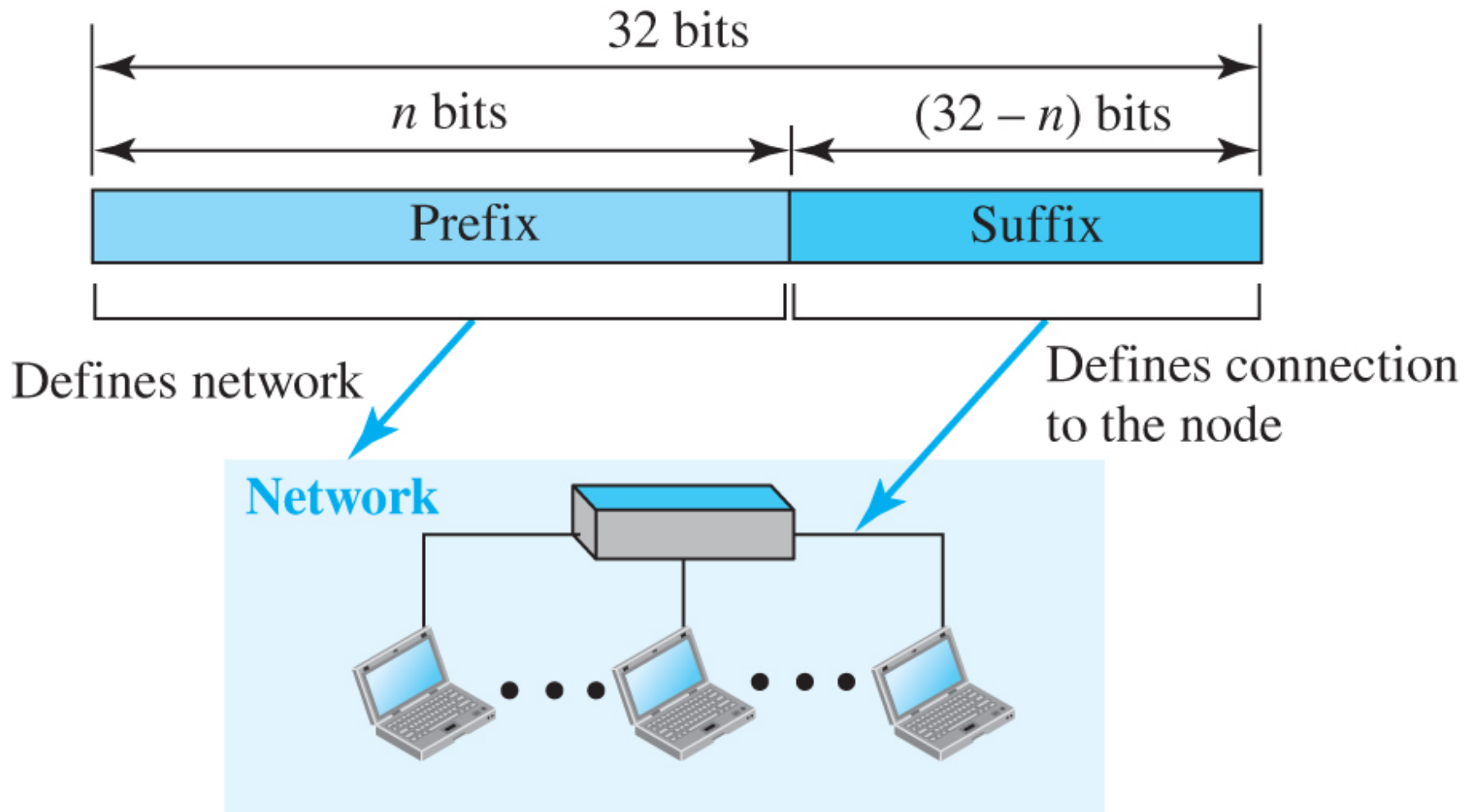


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Hierarchy in Addressing

- *In any communication network that involves delivery, such as a telephone network or a postal network, the addressing system is hierarchical.*
- *A 32-bit IPv4 address is also hierarchical but divided only into two parts.*
- *The first part of the address, called the prefix, defines the network;*
- *the second part of the address, called the suffix, defines the node.*

Figure 7.5 Hierarchy in addressing

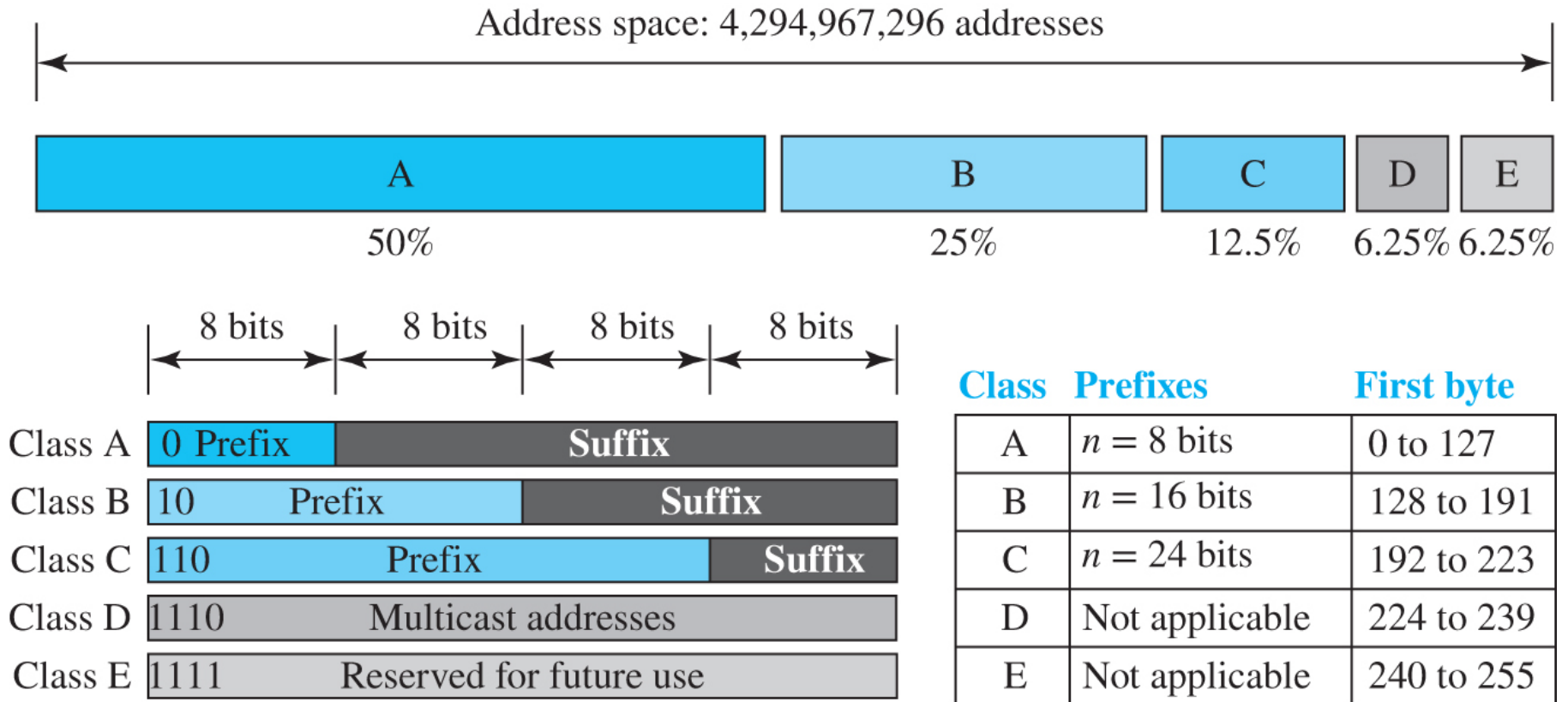


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Classful Addressing

- *When the Internet started, an IPv4 address was designed with a fixed-length prefix, but to accommodate both small and large networks, three fixed-length prefixes were designed instead of one ($n = 8$, $n = 16$, and $n = 24$).*
- *The whole address space was divided into five classes (class A, B, C, D, and E), as shown in Figure 7.6.*
- *This scheme is referred to as classful addressing.*
- *Although classful addressing belongs to the past, it helps us to understand classless addressing, discussed later.*

Figure 7.6 Occupation of the address space in classful addressing



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Example

Find the class of each address.

- a.* 00000001 00001011 00001011 11101111
- b.* 11000001 10000011 00011011 11111111
- c.* 14.23.120.8
- d.* 252.5.15.111

Solution

- a.* *The first bit is 0. This is a class A address.*
- b.* *The first 2 bits are 1; the third bit is 0. This is a class C address.*
- c.* *The first byte is 14; the class is A.*
- d.* *The first byte is 252; the class is E.*

Note

In classful addressing, a large part of the available addresses were wasted.

Table *Default masks for classful addressing*

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255 .0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255 .0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255 .0	/24

Note

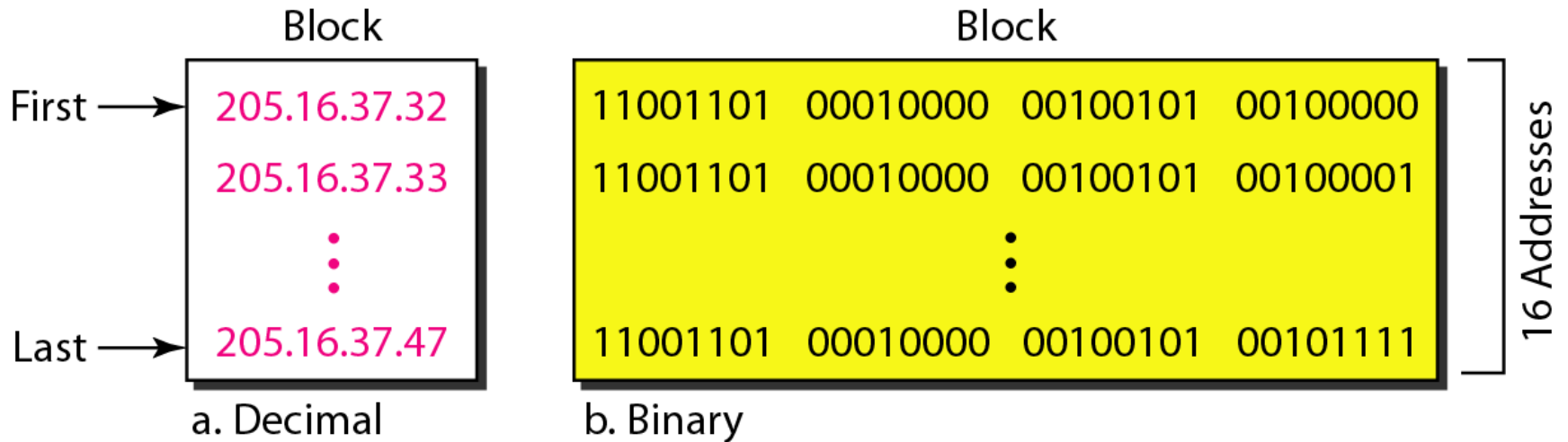
Classful addressing, which is almost obsolete, is replaced with classless addressing.

Example

Figure 7.7 shows a block of addresses, in both binary and dotted-decimal notation, granted to a small business that needs 16 addresses.

We can see that the restrictions are applied to this block. The addresses are contiguous. The number of addresses is a power of 2 ($16 = 2^4$), and the first address is divisible by 16. The first address, when converted to a decimal number, is 3,440,387,360, which when divided by 16 results in 215,024,210.

Figure 7.7 *A block of 16 addresses granted to a small organization*



Note

The first address in the block can be found by setting the rightmost $32 - n$ bits to 0s.

The last address in the block can be found by setting the rightmost $32 - n$ bits to 1s.

The number of addresses in the block can be found by using the formula 2^{32-n} .

Note

**Each address in the block can be considered as a two-level hierarchical structure:
the leftmost n bits (prefix) define the network;
the rightmost $32 - n$ bits define the host.**

Figure 7.8 *Configuration and addresses in a subnetted network*

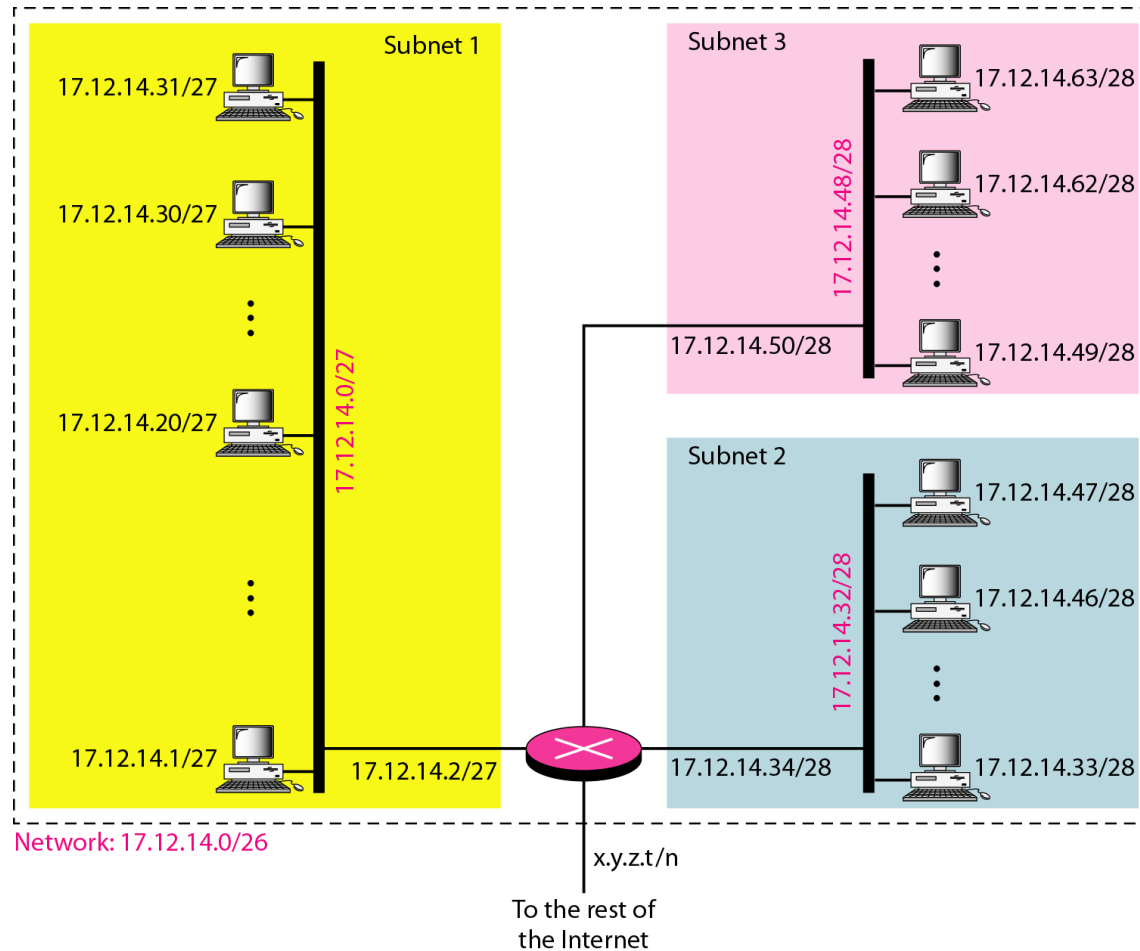


Figure 7.9 *Three-level hierarchy in an IPv4 address*

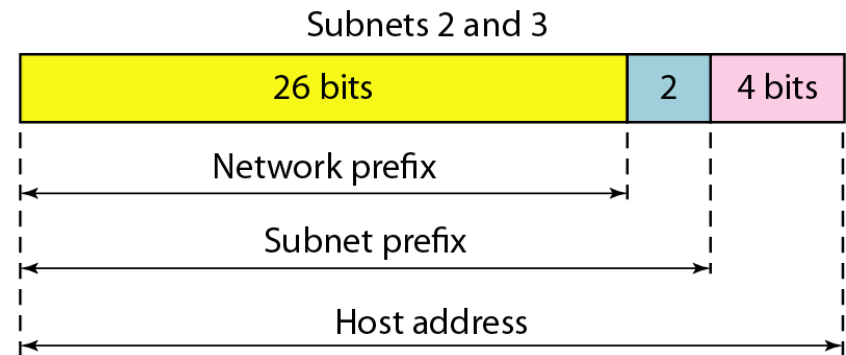
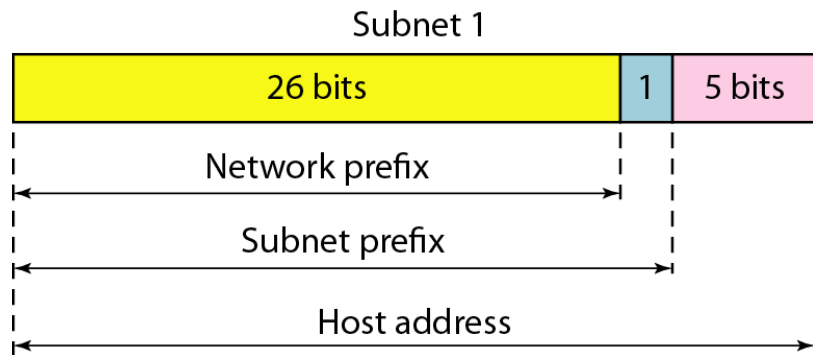


Figure 7.10 *An example of address allocation and distribution by an ISP*

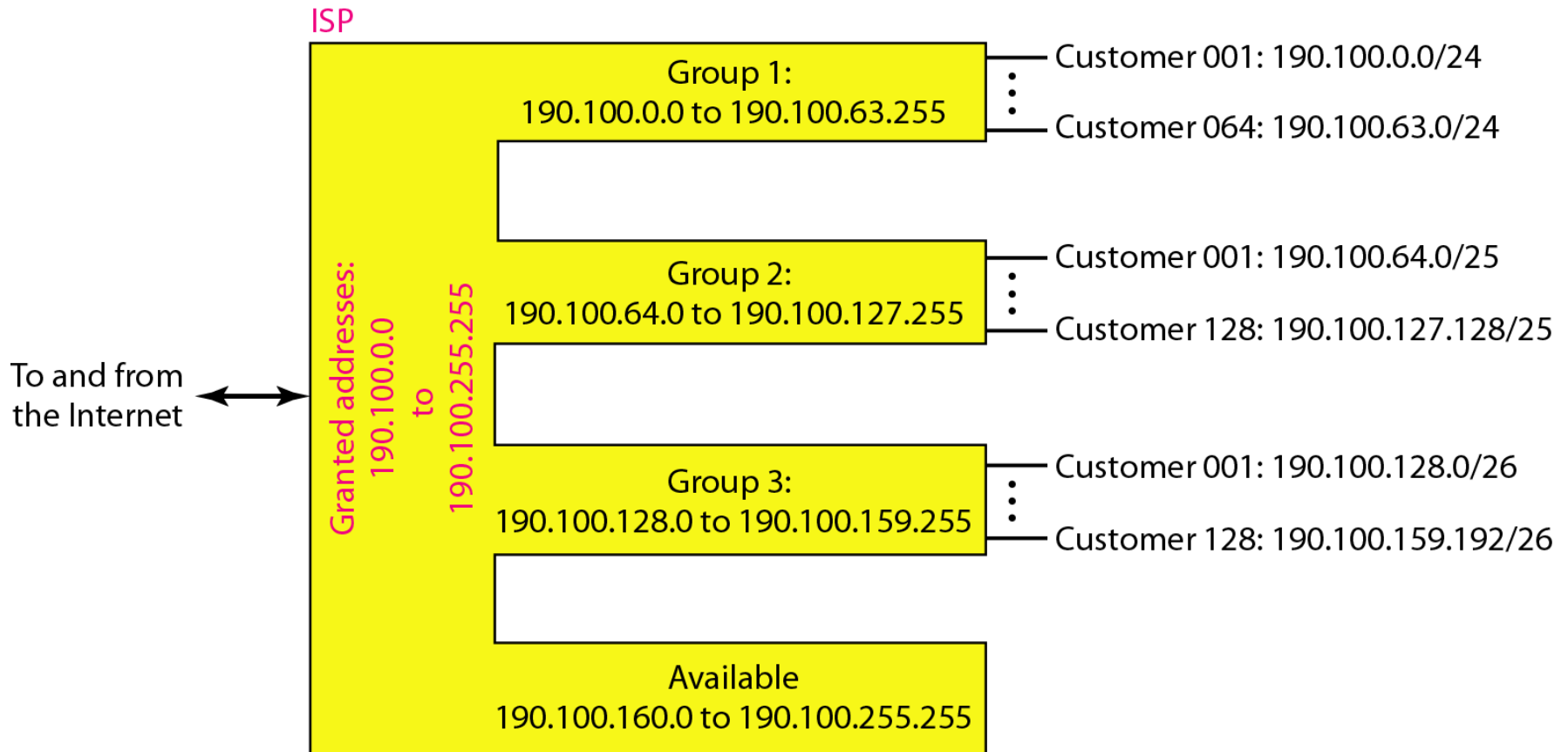


Table *Addresses for private networks*

<i>Range</i>			<i>Total</i>
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	172.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}

Figure 7.11 *A NAT implementation*

Site using private addresses

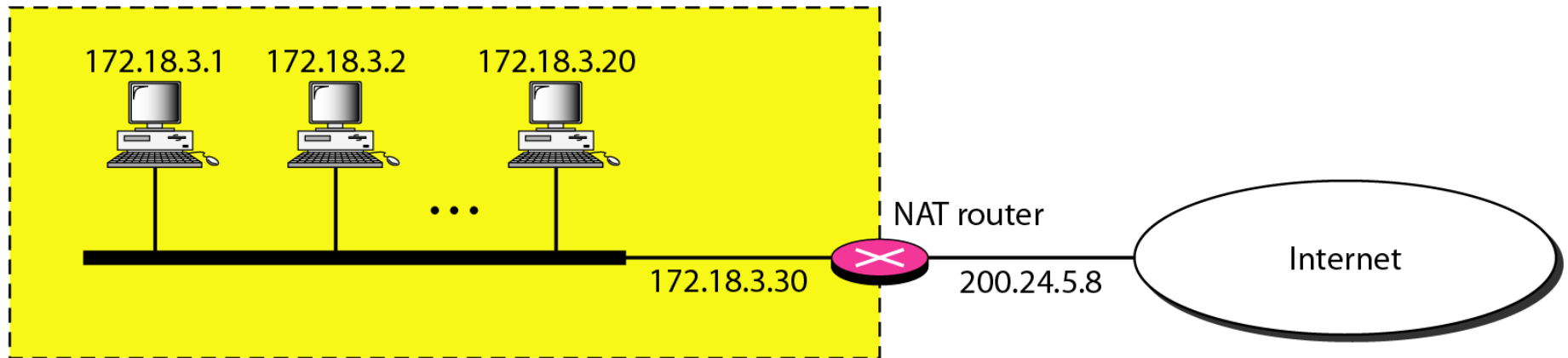


Figure 7.12 *Addresses in a NAT*

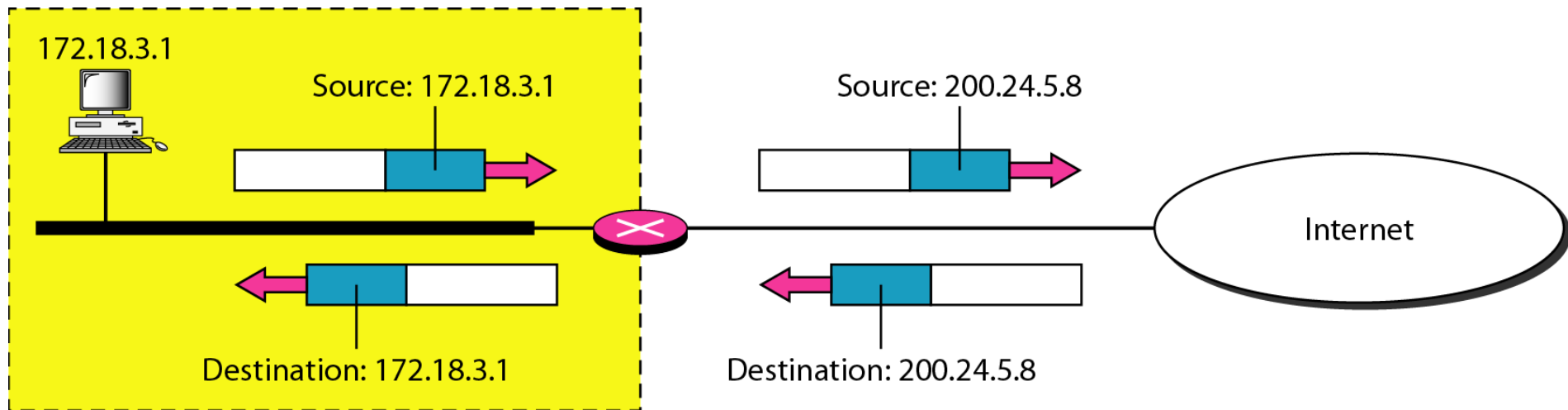


Figure 7.13 *NAT address translation*

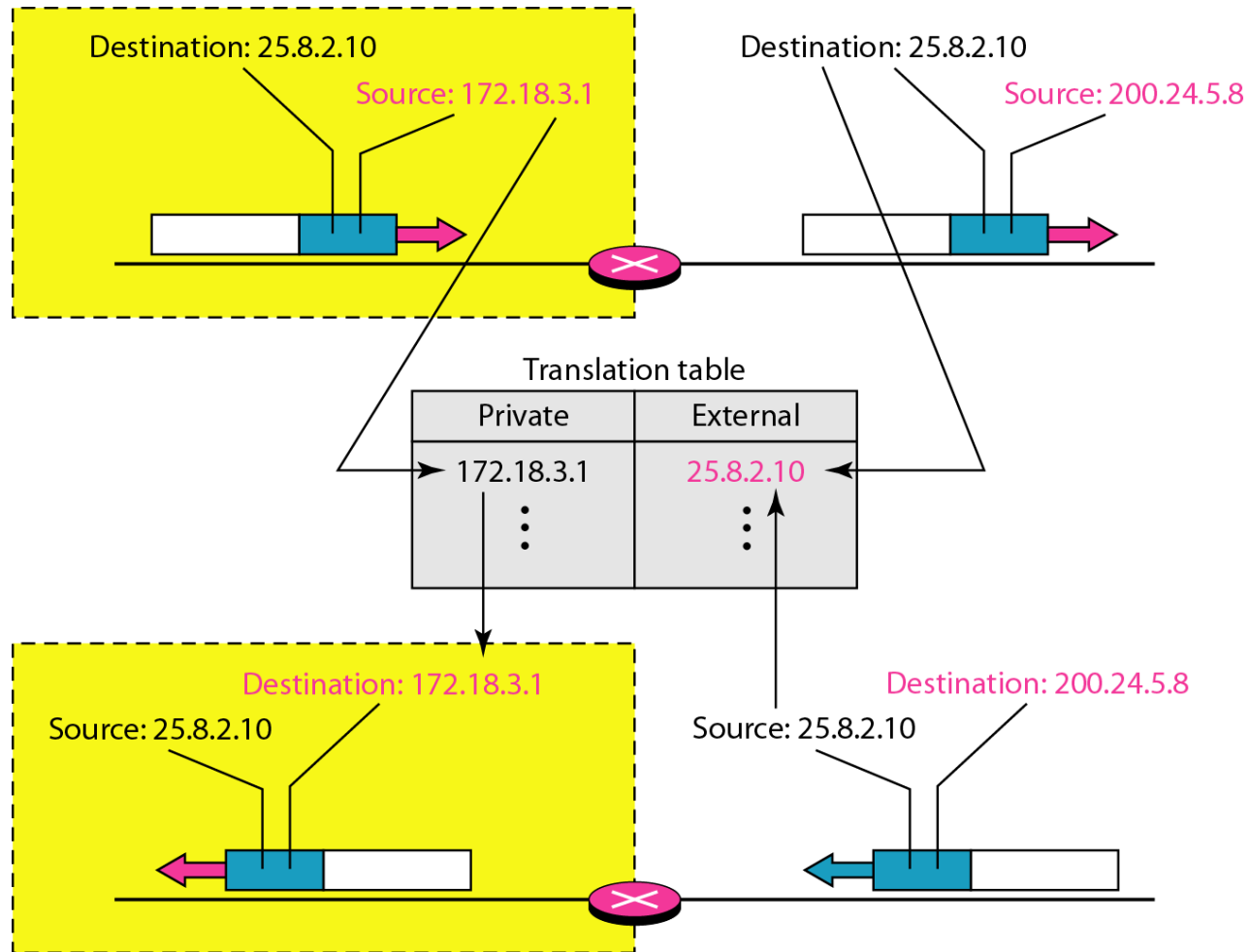
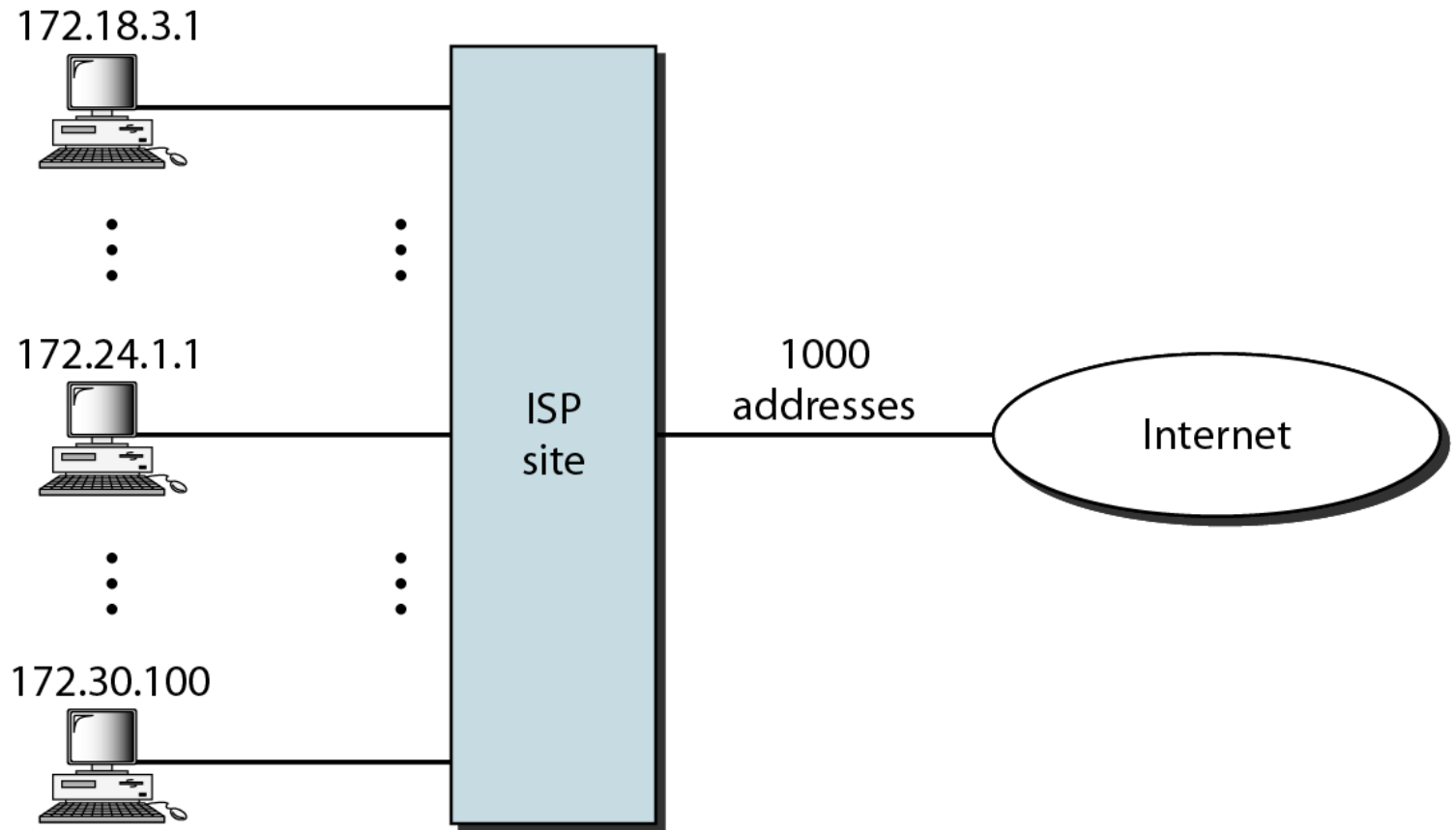


Table 7.14 *Five-column translation table*

<i>Private Address</i>	<i>Private Port</i>	<i>External Address</i>	<i>External Port</i>	<i>Transport Protocol</i>
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
...	...	...	...	...

Figure 7.15 *An ISP and NAT*



Security of IPv4 Datagrams

We give a brief idea about the security issues in IP protocol.

Packet Sniffing

An intruder may intercept an IP packet and make a copy of it.

Packet Modification

The attacker intercepts the packet, changes its contents, and sends the new packet to the receiver..

IP Spoofing

An attacker can masquerade as somebody else and create an IP packet that carries the source address of another computer.

IPSec

The IP packets today can be protected from the previously mentioned attacks using a protocol called IPSec (IP Security). This protocol, which is used in conjunction with the IP protocol, creates a connection-oriented service between two entities in which they can exchange IP packets without worrying about the three attacks discussed above. We will discuss IPSec in detail in Chapter 13.

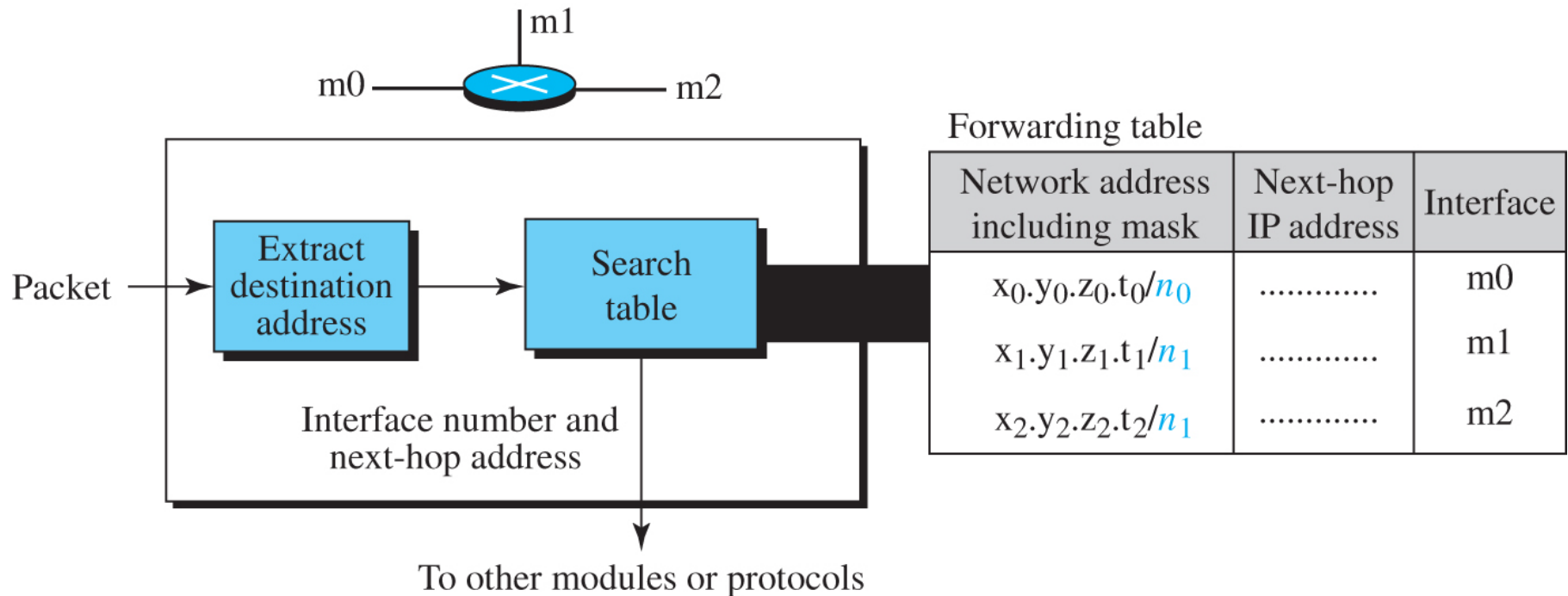
7.4.6 Forwarding of IP Packets

- *We discussed the concept of forwarding at the network layer earlier in this chapter.*
- *In this section, we extend the concept to include the role of IP addresses in forwarding.*
- *As we discussed before, forwarding means to place the packet in its route to its destination.*
- *Since the Internet today is made of a combination of links (networks), forwarding means to deliver the packet to the next hop (which can be the final destination or the intermediate connecting device).*
- *Although the IP protocol was originally designed as a connectionless protocol, today the tendency is to change it to connection-oriented protocol. We discuss both cases.*

Forwarding Based on Destination Address

- *We first discuss forwarding based on the destination address.*
- *This is a traditional approach, which is prevalent today.*
- *In this case, forwarding requires a host or a router to have a forwarding table.*
- *When a host has a packet to send or when a router has received a packet to be forwarded, it looks at this table to find the next hop to deliver the packet to.*

Figure 7.32 Simplified forwarding module in classless address

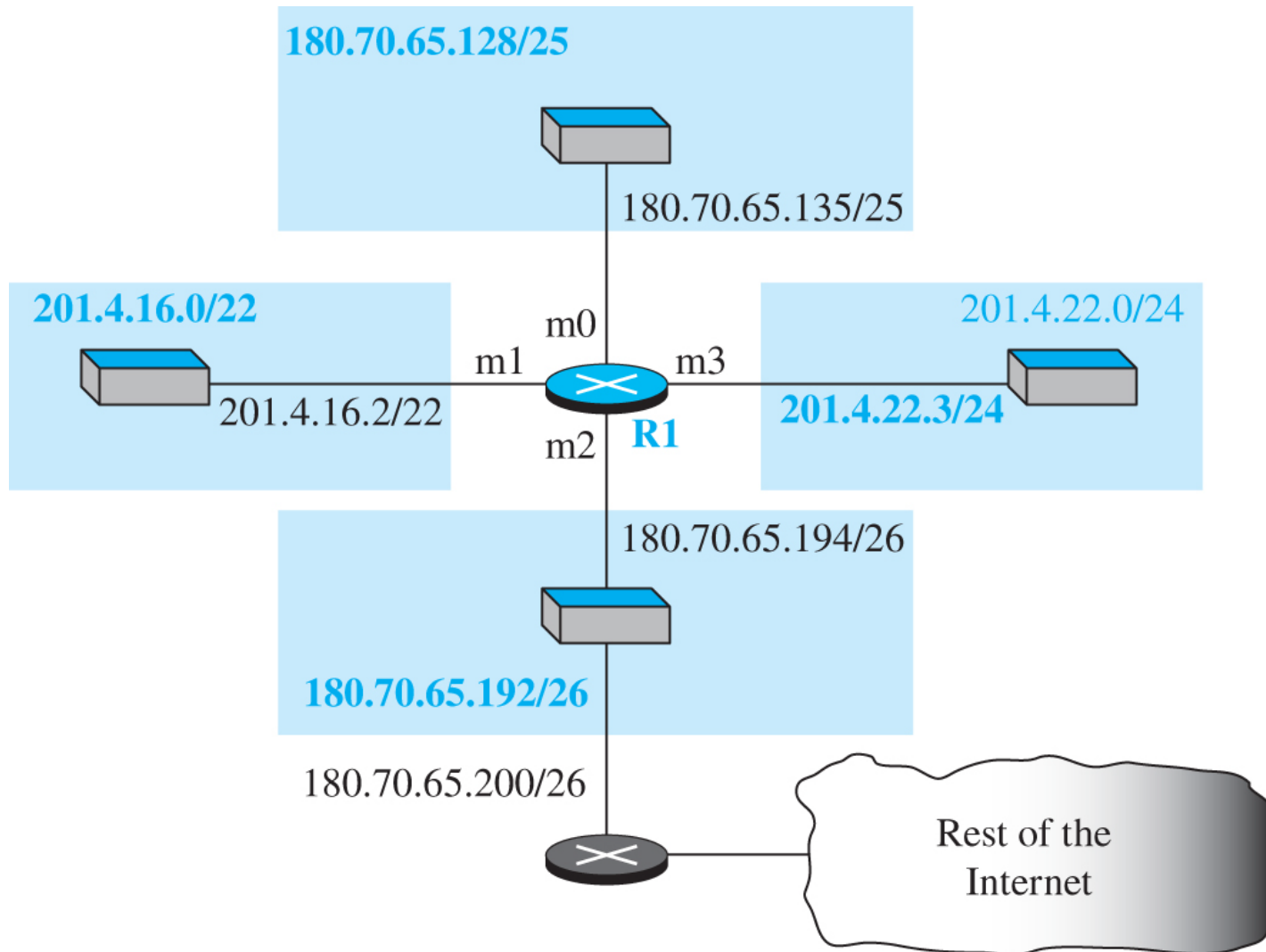


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Example 7.19

Make a forwarding table for router R1 using the configuration in Figure 7.33.

Figure 7.33 Configuration for Example 7.19



[Access the text alternative for slide images.](#)

Table 7.3 Forwarding table for router R1

Network address/mask	Next hop	Interface
180.70.65.192/ 26	—	m2
180.70.65.128/ 25	—	m0
201.4.22.0/ 24	—	m3
201.4.16.0/ 22	—	m1
Default	180.70.65.200	m2

Example 7.20

Instead of Table 7.3, we can use Table 7.4, in which the network address/mask is given in bits.

Table 7.4 Forwarding table for router R1 using prefix bit

<i>Leftmost bits in the destination address</i>	<i>Next hop</i>	<i>Interface</i>
10110100 01000110 01000001 11	—	m2
10110100 01000110 01000001 1	—	m0
11001001 00000100 00011100	—	m3
11001001 00000100 000100	—	m1
Default	180.70.65.200	m2

7.5 NEXT GENERATION IP (IPv6)

The address depletion of IPv4 and other shortcomings of this protocol prompted a new version of IP protocol in the early 1990s, which is called Internet Protocol version 6 (IPv6) or IP new generation (Ipng).

7.5.1 IPv6 Addressing

The main reason for migration from IPv4 to IPv6 was the small size of the address space of IPv4. An IPv6 address is 128 bytes or 16 bytes, four times the address length in IPv4.

Why Change IP version?

Address space exhaustion

- *two level addressing (network and host) wastes space*
- *network addresses used even if not connected*
- *growth of networks and the Internet*
- *extended use of TCP/IP*
- *single address per host*
- *requirements for new types of service*

Representation

An IPv6 address is 128 bits or 16 bytes long; four times the address length of IPv4.

Binary (128 bits)

1111110111101101011 ... 111111100000000

Colon hexadecimal

FEF6:BA98:7654:3210:ADEF:BBFF:2922:FF00

Address Space₂

The address space of IPv6 contains 2^{128} addresses. This address space is 2^{96} times the IPv4 address—definitely no address depletion—as shown, the size of the space is

340, 282, 366, 920, 938, 463, 374, 607, 431, 768, 211, 456

Figure 7.34 *IPv6 address in binary and hexadecimal colon notation*

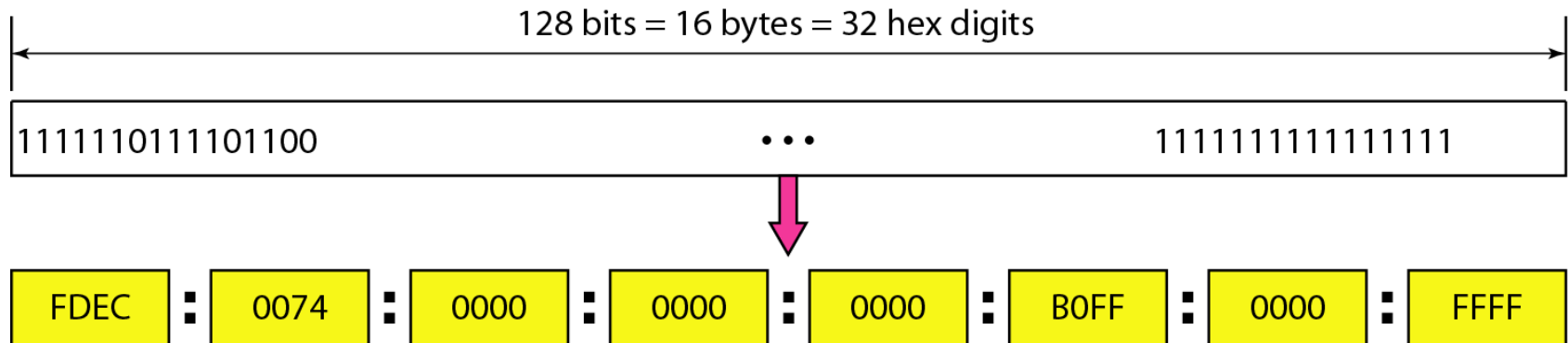
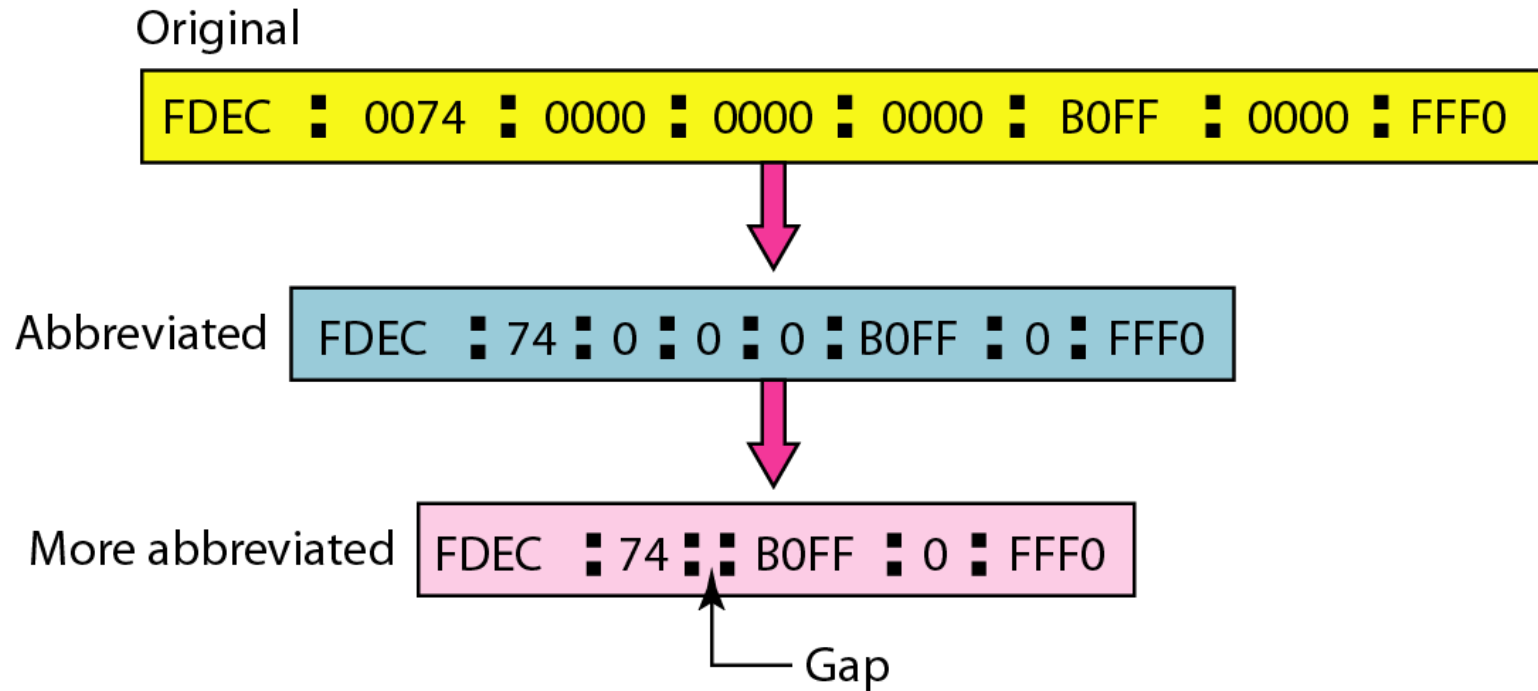


Figure 7.35 *Abbreviated IPv6 addresses*



Expand the address 0:15::1:12:1213 to its original.

Solution

We first need to align the left side of the double colon to the left of the original pattern and the right side of the double colon to the right of the original pattern to find how many 0s we need to replace the double colon.

XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX
0: 15: : 1: 12:1213

This means that the original address is.

0000:0015:0000:0000:0000:0001:0012:1213

7-6 TRANSITION FROM IPv4 TO IPv6

Although we have a new version of the IP protocol, how can we make the transition to stop using IPv4 and start using IPv6? in the Internet can move The transition must be smooth to prevent any problems between IPv4 and IPv6 systems.

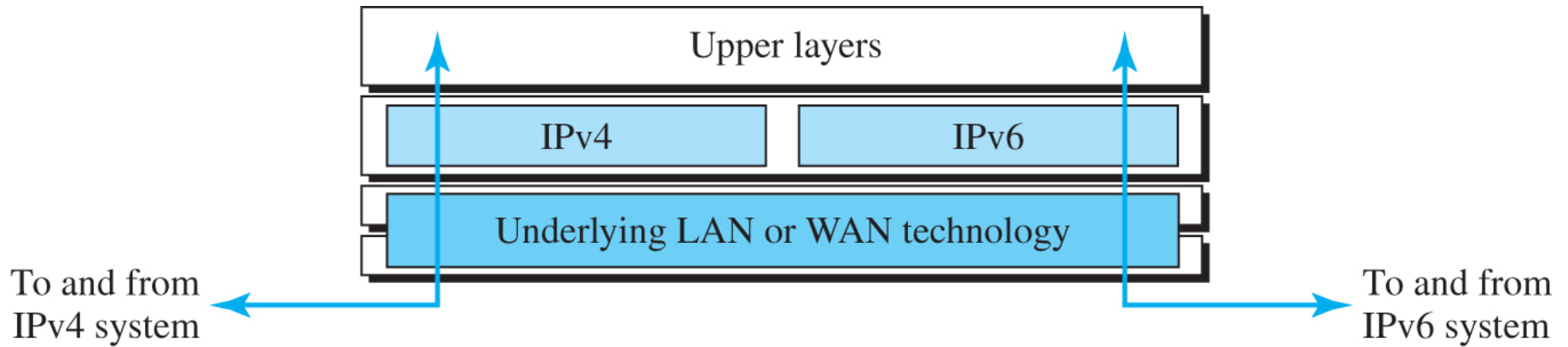
7.6.1 Strategies

Three strategies have been devised for transition: dual stack, tunneling, and header translation. One or all of these three strategies can be implemented during the transition period.

Dual Stack

It is recommended that all hosts, before migrating completely to version 6, have a dual stack of protocols during the transition. In other words, a station must run IPv4 and IPv6 simultaneously until all the Internet uses IPv6. See Figure 7.51 for the layout of a dual-stack configuration

Figure 7.51 Dual stack

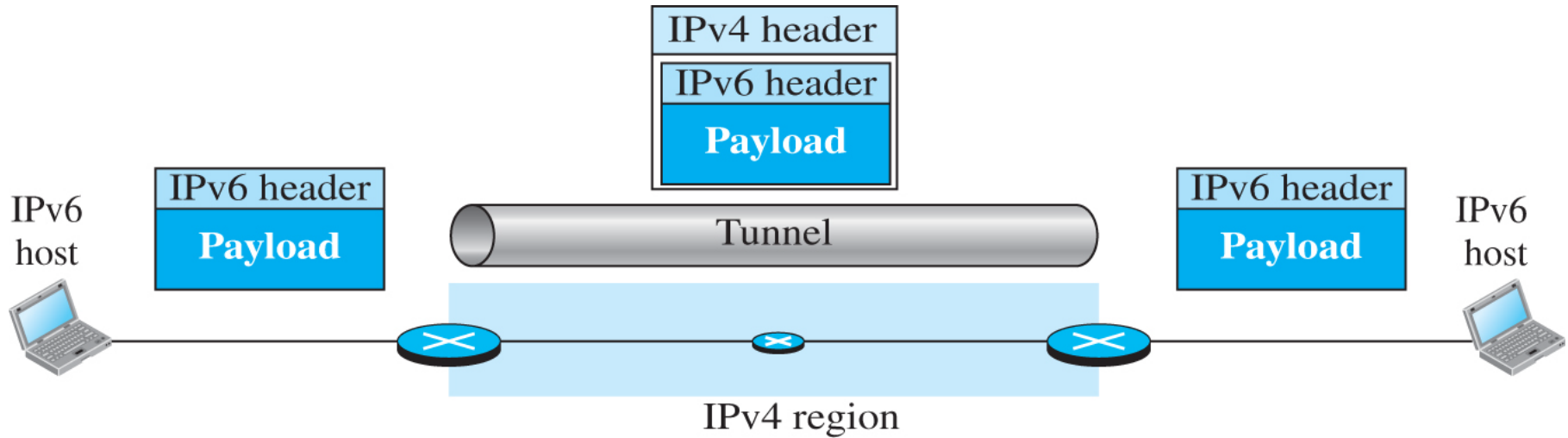


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Tunneling

Tunneling is a strategy used when two computers using IPv6 want to communicate with each other and the packet must pass through a region that uses IPv4. To pass through this region, the packet must have an IPv4 address. So the IPv6 packet is encapsulated in an IPv4 packet when it enters the region, and it leaves its capsule when it exits the region. It seems as if the IPv6 packet goes through a tunnel at one end and emerges at the other end. To make it clear that the IPv4 packet is carrying an IPv6 packet as data, the protocol value is set to 41. Tunneling is shown in Figure 7.52.

Figure 7.52 Tunneling strategy

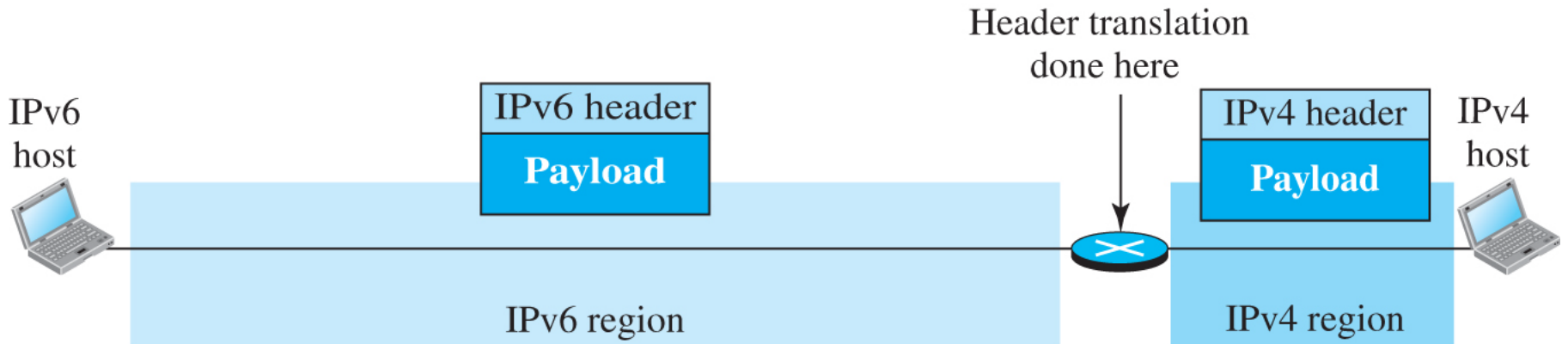


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Header Translation

Header translation is necessary when the majority of the Internet has moved to IPv6 but some systems still use IPv4. The sender wants to use IPv6, but the receiver does not understand IPv6. Tunneling does not work in this situation because the packet must be in the IPv4 format to be understood by the receiver. In this case, the header format must be totally changed through header translation. The header of the IPv6 packet is converted to an IPv4 header (see Figure 7.53).

Figure 7.53 Header translation strategy



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